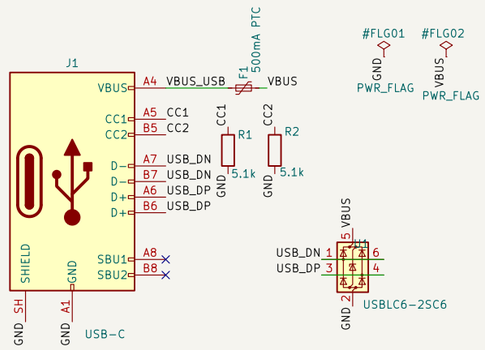
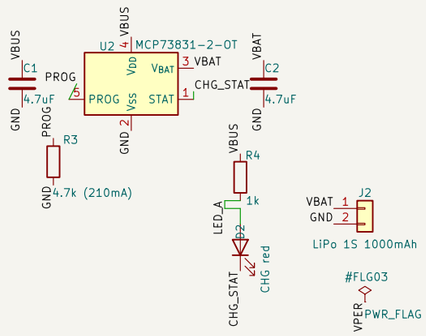
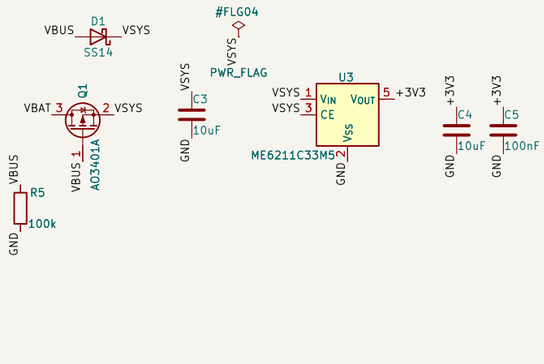
USB-C INPUT + ESD (CC 5.1k = 5V sink)



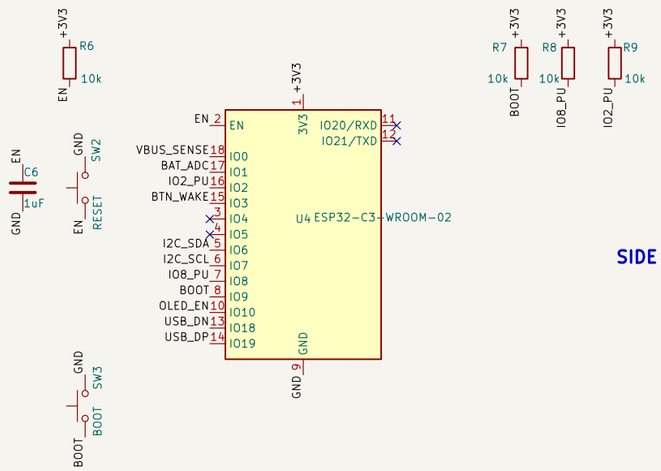
LiPo CHARGER 210mA (cool)



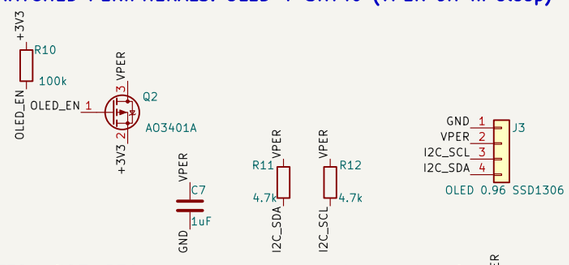
POWER PATH (USB -> load, battery isolated) + 3V3 LDO



MCU ESP32-C3 (native USB: Arduino IDE, no USB-UART chip)



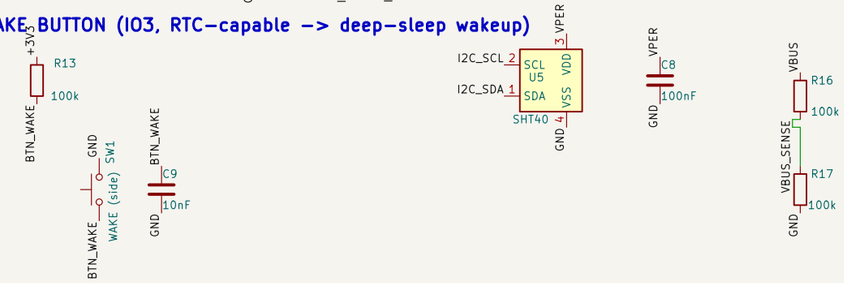
SWITCHED PERIPHERALS: OLED + SHT40 (VPER off in sleep)



SENSING: VBAT/2 -> IO1, VBUS/2 -> IO0



SIDE WAKE BUTTON (IO3, RTC-capable -> deep-sleep wakeup)



SHT40 on thermally isolated PCB tongue, OLED 0.96 SSD1306 I2C on top
LiPo 1S 1000mAh, MCP73831 210mA charger, power-path, ME6211 LDO

Sheet: /
File: ThermoBox.kicad_sch

Title: ThermoBox – ESP32-C3 thermometer / one-button game box

Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. 10.0.8		Id: 1/1